

Elicitation

1. What kind of extra administration controls would you want for this kind of app? E.g. More ability to change user permissions, set roles.
2. What kind of file sharing capabilities would you suggest to incorporate into the app?
3. Would you like to see any collaborative document or brainstorm features on the app? If so, do you have any examples?
4. What other communication tools would you like to see on the flockr?
5. Would you like to see some sort of to do list or a way to keep track of progress? If so what would you personally like to see that you prefer or haven't seen done before?
6. What other extra features would you expect an app like Flockr to have as a teamwork-driven communication tool?

Lachlan (lachlan1288@gmail.com)

1. No point in having extra administration controls in a student group project learning environment. If this were to be used in terms of teaching and classwork then yes.
2. Anything that you can upload that would be important for learning
 - a. PDFs, word documents, images, videos
3. Collaborative tools like google jamboards
 - a. Google jamboards - Put sticky notes and contribute
 - b. Google docs - Live editing and writing
4. I would like to directly message people. This would allow members to discreetly give others feedback without causing any undue embarrassment. Calling would be a must-have feature. In my mind, there is no point in using it without it.
5. Yes I would like to keep everyone on track by using a kind of virtual notice board. I would like to set due dates for tasks, tick stuff off to track progress and assign tasks to people
6. I can't really think about anything

Joanne (z5257044@ad.unsw.edu.au)

1. The ability to kick someone out of a channel or the server could be useful.
2. PDFs, documents, images. The more the better.
3. An imitation google drive where everyone can share files instead of sending things one at a time. An imitation google doc where everyone can work on a file at the same time, a collaborative whiteboard to jot things down during meetings and a polls system would be useful for making decisions as a group.
4. Direct messages are a must for contacting people individually. A calling capability would be really useful for organising meetings
5. I would like to see a to do list where teams can note things to be done with due dates and who the task is assigned to. It would be nice if you could link this to calendar or receive notifications on your phone about when something that needs to be done is due soon.
6. Notification reminders are also needed to keep the team driven. The ability to delete a channel or a user would also be useful.

Analysis & Specification - Use Cases

As a student/flockr user, I want to be able to kick people out of channels, so that if a member accidentally invited someone who shouldn't be in the channel, they should be able to remove them.

Acceptance Criteria

- Next to the button to add/remove a channel member as a channel owner, there will be a new option to kick out the member.
- The button icon will be a boot
- Once a channel owner clicks this button, the selected member will be removed from the channel.
- If a member who is not a channel owner presses this button, an error will be raised informing them they do not have the authority to do this.
- Channel owners cannot kick themselves out, there will be no button available to them.

Use Case	Kick user
Primary Actor	Channel owner
Goal in Context	Remove member who accidentally joined or a a disruptive member
Scope	Channel, Flockr backend
Trigger	Channel owner presses the kick user button
Preconditions	• Person removing the member must be a channel owner. • The user to be kicked joined the channel by mistake, was accidentally invited, or is disrupting the team.

Basic flow	<u>MAIN SUCCESS SCENARIO</u> 1. Owner looks for user they would like to kick in the list of channel members 2. Owner clicks on the kick button next to the users name 3. Backend checks whether the requested user is part of the channel and whether the teacher is a channel owner 4. The specified user is removed from the member list on the channel
End Condition	The specified user is removed from the channel
Failed End Condition	The authorised user is not a channel owner

Comments:

- This addresses the problems I had if Flockr were to be used in a teaching/classroom environment. (Lachlan)
- Although, I think the ability to ban someone from a channel would be useful to prevent unwanted users from continually joining the channel, whether it be by accident or because they think they're funny. (Lachlan)
- This matches what I had in mind when I suggested a kick member feature. (Joanne)

As a student, I would like to upload a range of files and send them as a message so that my team can share resources for learning.

Acceptance Criteria

- In addition to the 'send later' icon in the message dialog box, there will be a 'plus' icon for uploading/attaching files to a message.
- Clicking on the icon will open a dialog box prompting the user to choose file(s) to upload (video, image, pdf etc.)
- Once the user selects the file(s), they will become attached to the message.
- The user can send the message as it is, sending only the files, or can write a message to go along with the files.
- Users will be able to view the files sent in the message and can download them

Use Case	Upload file and send as message
Primary Actor	Flockr user
Goal in Context	User wants to upload a file and send it as message in the channel to share it with other members
Scope	Channel, Flockr backend
Trigger	User pressed the upload/attach file button
Preconditions	● User is a member of the channel ● The file they selected to upload is of the supported type
Basic flow	<u>MAIN SUCCESS SCENARIO</u> 1. User has a file they wish to upload and send as a message 2. User clicks the upload/attach file button on the message bar 3. User selects the file(s) they wish to upload 4. Backend checks whether the files are of the support type 5. The file(s) are attached to the message 6. User can choose to attach a message 7. User sends the message 8. Message is sent to channel with the user given text and file(s)

End Condition	A message is sent to the channel containing the file(s) which can be viewed by other channel members
Failed End Condition	The file type is not supported

Comments:

- I think this addresses the problem I had with the lack of file sharing capabilities on the Flockr very well. (Lachlan)
- This is the kind of feature I had in mind but I also would like a kind of shared drive that users could upload files to and anyone could access (similar to Google Drive). (Lachlan)
- This is a good and reasonable way to incorporate file sharing. I think this would be good enough for team use but I still would like to see a imitation google drive where we can dump files in the future. (Joanne)

As a student, I would like to directly message people so that I can contact teachers privately or members in my group for feedback.

Acceptance Criteria

- New tab on the side to view all members of the flockr in a list. List is divided into two categories: Flockr owners and regular members
- Can click on any member on the list and select the option to message them directly.
- Messaging someone directly will open a messages interface similar to that already in the channels
- Direct message history will be saved on a list that can be accessed by clicking on a new tab on the side bar: 'Private messages'
- Similar to facebook, these will be a list of conversations with different users that the user can click on to re-open the message interface and send messages to that person.

Use Case	Direct message a certain user
Primary Actor	Flockr user
Goal in Context	User wants to send a message only to a certain user
Scope	Channel, Flockr backend
Trigger	User chooses a member and selects the option to message them directly.
Preconditions	Chosen user has a valid user id
Basic flow	<u>MAIN SUCCESS SCENARIO</u> 1. Choose direct message option to open interface 2. User selects someone from the list of flockr members to message 3. User sends message 4. Message is saved in the conversation with that person, which can be later accessed by opening the private messages tab.

End Condition	A message is sent to the chosen user. This can only be viewed and replied to by that user.
Failed End Condition	Message is longer than 1000 characters

Comments:

- This is good. I don't really have any issues/feedback for this particular user story. (Lachlan)
- Does the list of members need to be divided into owner members and other members? I feel like it would be fine to just have a list of all members and choose a member to message from there. (Joanne)

As a student working on a project with a group, I would like a to-do-list/task board feature because it would allow us to keep track of our progress on the project as a group and quickly respond to any problems that may arise.

Acceptance Criteria

- Within a channel, there will be a new option called 'task board'
- Clicking on it will open the task board page
- The board is divided into three columns: 'To Do', 'Doing', 'Completed'
- Users are able to add new tasks to the 'To Do' section by pressing the (+) button at the top of the board.
- A pop up window will prompt the user to write down the task, set its due date and either assign it to themselves or someone else
- The newly created task will then appear on the list of tasks in 'To Do', stylised to have the text contained within a box or sticky note like shape
- Channel owners and task assignees can click and drag the task 'sticky notes' to the other two columns or remove it entirely.
- When a task due date is close and the task is still in the 'To Do' section, a red exclamation mark symbol will appear next to the 'task board' option in the channel and the overdue task will be highlighted on the board itself.
- The symbol/highlight remains even past the due date, until the assignee moves it

Use Case	Add a task to task board and change status
Primary Actor	Flickr user
Goal in Context	User adds a task to the task board, assigns it to another user and sets it to 'doing'
Scope	Channel, Flickr backend
Trigger	User selects task board option in channel
Preconditions	• Authorised user is a member of the channel • Target user is part of the channel

Basic flow	<u>MAIN SUCCESS SCENARIO</u> 1. User clicks the task board option in channel 2. User clicks the (+) button at the top of the board 3. Add task description: "Finish introduction for report" 4. Assign the task to a user in the channel, Dave 5. Add a due date 6. Backend checks whether provided parameters are valid 7. Task appears on the task board
End Condition	Task added to the task board with 'to do' label and is assigned to Dave
Failed End Condition	• Task description is longer than 1000 characters • Due date is some time in the past • Assignee is not a valid user • There are more than 200 tasks

Comments:

- This is good. I don't really have any issues/feedback for this particular user story. (Lachlan)
- This looks good, will definitely be useful for planning teamwork. (Joanne)

As a student, I want calling features on the app because it allows for more direct and clear communication with teachers and other students. It is an essential feature.

Acceptance Criteria

- Next to the invite member, leave channel options, there will be a new option: 'Start call'. Any channel member is able to do this
- Once pressed, a phone ringing tone will play and a popup will appear for all members in the channel, detailing which channel the call is in and asking channel members if they wish to accept or decline the call.
- The popup will appear to any channel member regardless of what they are viewing in the flockr (i.e. even if they are not viewing the channel currently)
- Clicking accept will open a new browser window, where connected users are displayed in a list, people can speak through microphones and can leave the call by pressing the hangup button.
- If a user declined the call and wishes to join, they can navigate to the channel and click the 'Join call in progress' button that will be displayed at the top of the screen, in place of where 'Start call' would be.
- A call automatically ends when all connected members leave and a message will be sent in the chat saying: 'Call ended. Duration: <insert time>

Use Case	Start and end call in a channel
Primary Actor	Flockr user
Goal in Context	A call is started in a channel that any member can join, the call is then ended because everyone has left
Scope	Channel, Flockr backend
Trigger	Press start call option next to leave channel option
Preconditions	• User is a member of the channel • There are other members in the channel who will join the call and subsequently all leave

Basic flow	<u>MAIN SUCCESS SCENARIO</u> 1. Start call option is selected 2. A pop up appears for all members of the channel requesting for them to accept or decline the call 3. Other channel members join 4. If call is declined, call can be joined anytime by pressing 'Join call in progress' option 5. All users hang up from the call, causing the call to automatically end 6. A message is sent after the call has ended with the call duration
End Condition	Call starts successfully, users participate in it, before the call is ended.
Failed End Condition	Invalid channel

Comments:

- This is good. But I think the ability to share screens as I think it would be a really big help and a lot of other services have this feature. (Lachlan)
- This would make flockr viable against other team communications app. Perhaps consider adding a mute feature to the groups call (Joanne)

Interface Design

Function Name	HTTP Method	Parameters	Return type	Exceptions	Description
taskboard/view	GET	(token, channel_id)	{tasks}	InputError when any of: - Channel_id is invalid AccessError when - Authorised user is not a member of channel with channel_id	Given a Channel with ID channel_id that the authorised user is part of, open the task board for that channel and load its saved tasks. If any of the tasks are overdue, they will be marked as "overdue" and given special display treatment on the frontend
taskboard/add	POST	(token, channel_id, task, u_id, due_date)	{task_id}	InputError when any of: - Task description is longer than 1000 characters - Due date is in the past - There are already more than 200 tasks - u_id is not a member of the channel specified by channel_id AccessError when - Authorised user is not a member of channel with channel_id	Create a task in the task board of the channel specified by channel_id. Assign it to the user specified by u_id and set the given due date.

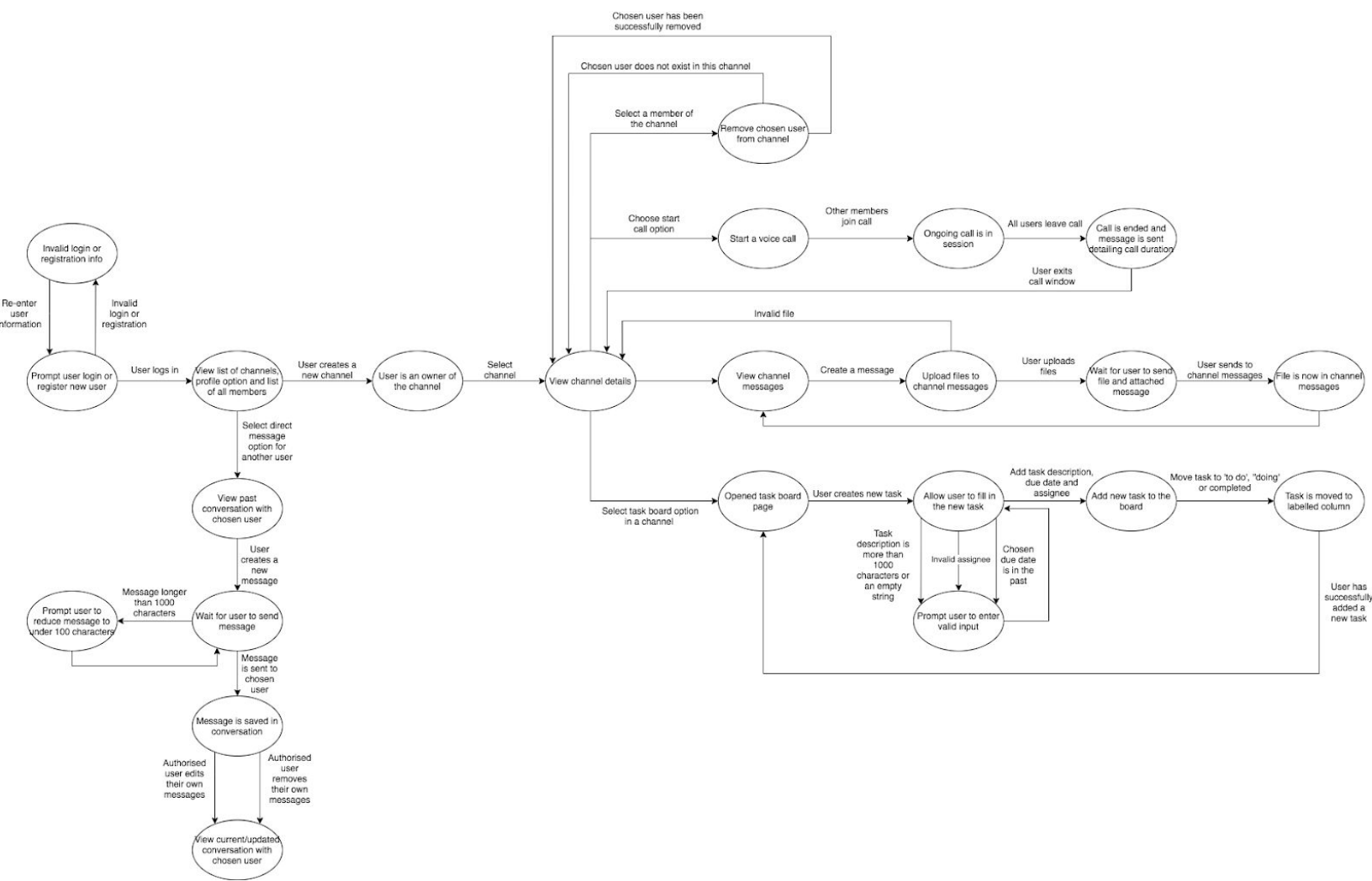
taskboard/remove	DELETE	(token, task_id)	{}	InputError when any of: - Task (based on id) does not exist AccessError when - Authorised user is not the task assignee or an owner of the channel specified by channel_id	Remove the task designated by task_id from the task board of the channel.
taskboard/edit	PUT	(token, task_id, task)	{}	InputError when any of: - Task description is more than 1000 characters AccessError when - Authorised user is not the task assignee or an owner of the channel specified by channel_id	Given a task, update it's text with new text. If the new task text is an empty string, the task is deleted.
taskboard/status	PUT	(token, task_id, status)	{}	InputError when any of: - Given status is not valid (not 1, 2 or 3) AccessError when - Authorised user is not the task assignee or an owner of the channel specified by channel_id	Given a task, update its status to one of three integers: 1 = To Do 2 = Doing 3 = Completed (Default value for newly created tasks is 1)

direct/message/send	POST	(token, u_id, message)	{u_id, message_id}	InputError when any of: - Specified u_id is invalid - Message is longer than 1000 characters	Send a message to user with u_id from authorised_user
direct/message/edit	PUT	(token, u_id, message_id, message)	{}	AccessError when: Message with message_id was not sent by the authorised user making this request	Given a direct message, update its text with new text. If the new message is an empty string, delete the message.
direct/message/remove	DELETE	(token, u_id, message_id)	{}	InputError when any of: - Message (based on ID) no longer exists AccessError when: - Message with message_id was not sent by the authorised user making this request	Remove a message sent directly to user with u_id
direct/message/view	GET	(token, u_id, start)	{messages, start, end}	InputError when any of: - u_id is not valid - Start is greater than number of messages in direct message history	Given a direct message chat with a user of u_id, return up to 50 messages between index "start" and "start + 50". Message with index 0 is the most recent message. Function returns a new index "end" which is the value of "start + 50", or, if this function has returned the least recent messages, returns -1 in "end" to indicate there are no more messages to load.

message/upload	POST	(token, channel_id, file)	{}	InputError when any of: - Channel ID is not a valid channel - Invalid file type AccessError when - Authorised user is not a member of channel with channel_id	Given a channel ID upload the given file as an attachment to a message
channel/kick	POST	(token, channel_id, u_id)	{}	InputError when any of: - Channel ID is not a valid channel - Specified u_id is invalid AccessError when - Authorised user is not an owner member of channel with channel_id	Given a channel ID and u_id, the specified user is removed as a member of this channel
call/start	POST	(token, channel_id)	{call_id}	InputError when any of: - Channel ID is not a valid channel AccessError when - Authorised user is not a member of channel with channel_id	Given a channel ID, start a call that other members can join
call/join	POST	(token, call_id)	{}	InputError when any of: Call ID is not a valid call	Add the authorised user to the call of call_id and its list of participants

call/leave	POST	(token, call_id)	{ }	InputError when any of: - Call ID is not a valid call AccessError when - Authorised user is not a participant of call with call_id	Given a call_id, remove the authorised user from that call and its list of participants
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State diagram



If its too blurry - We have attached a png of it in the project directory too